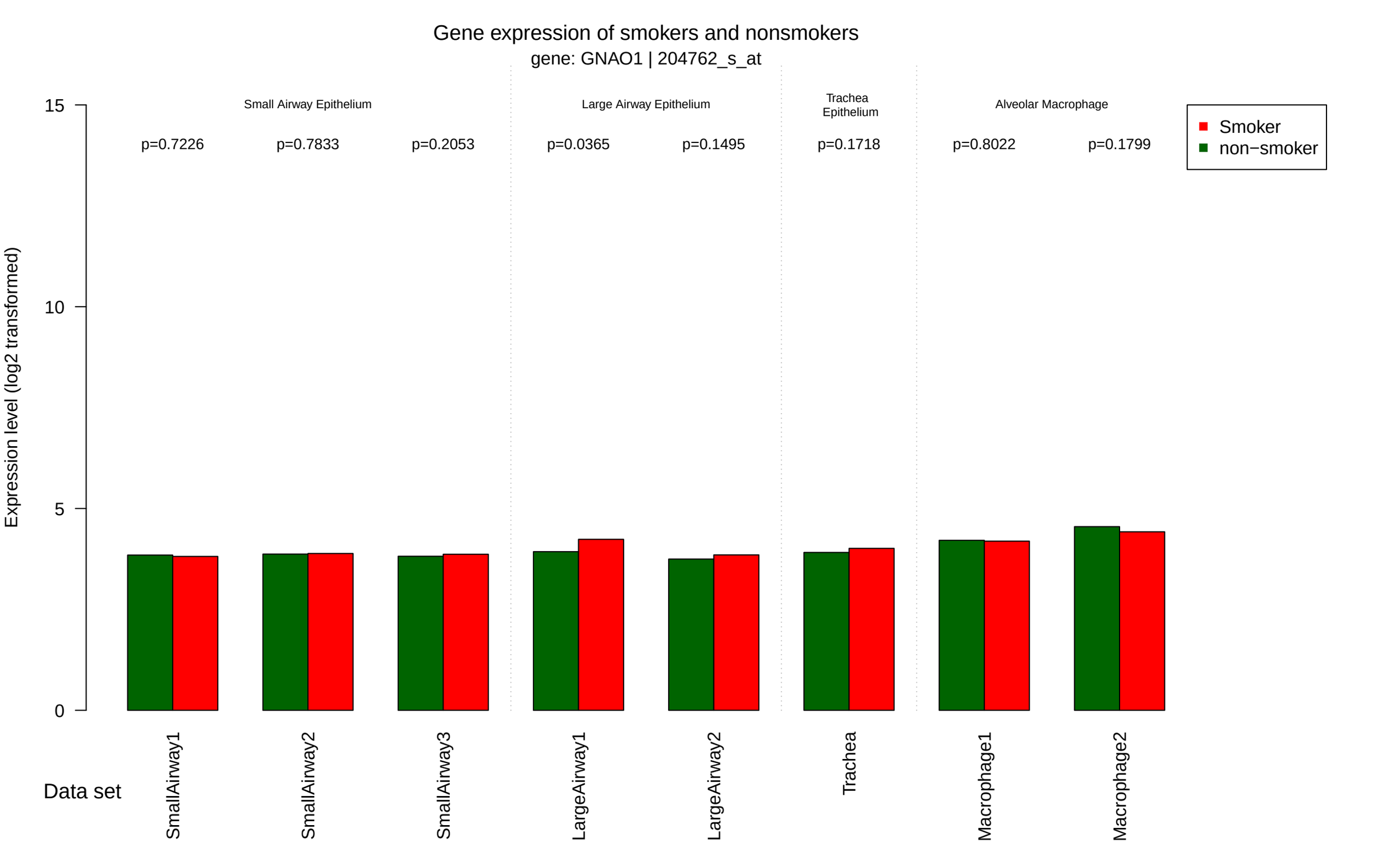
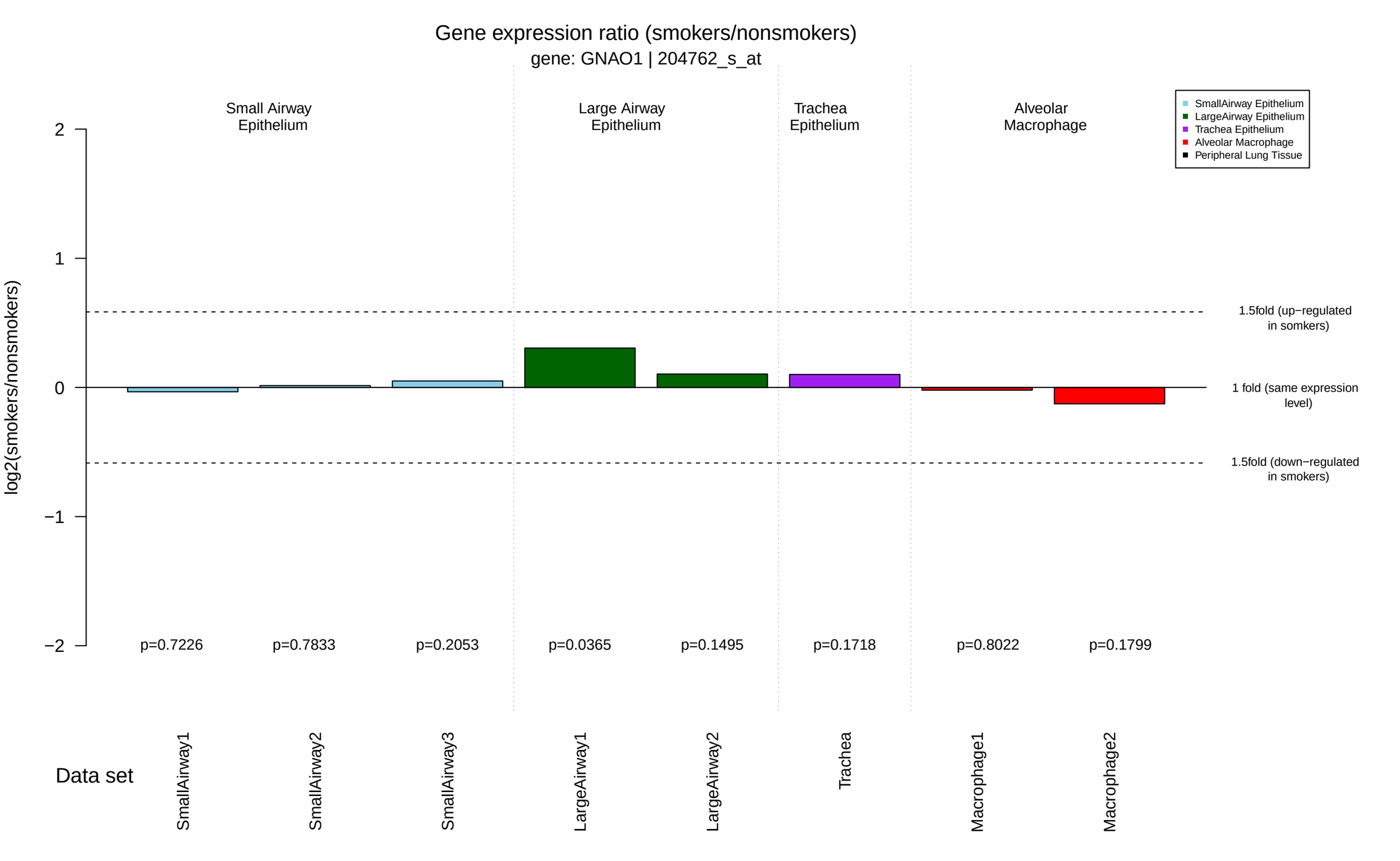
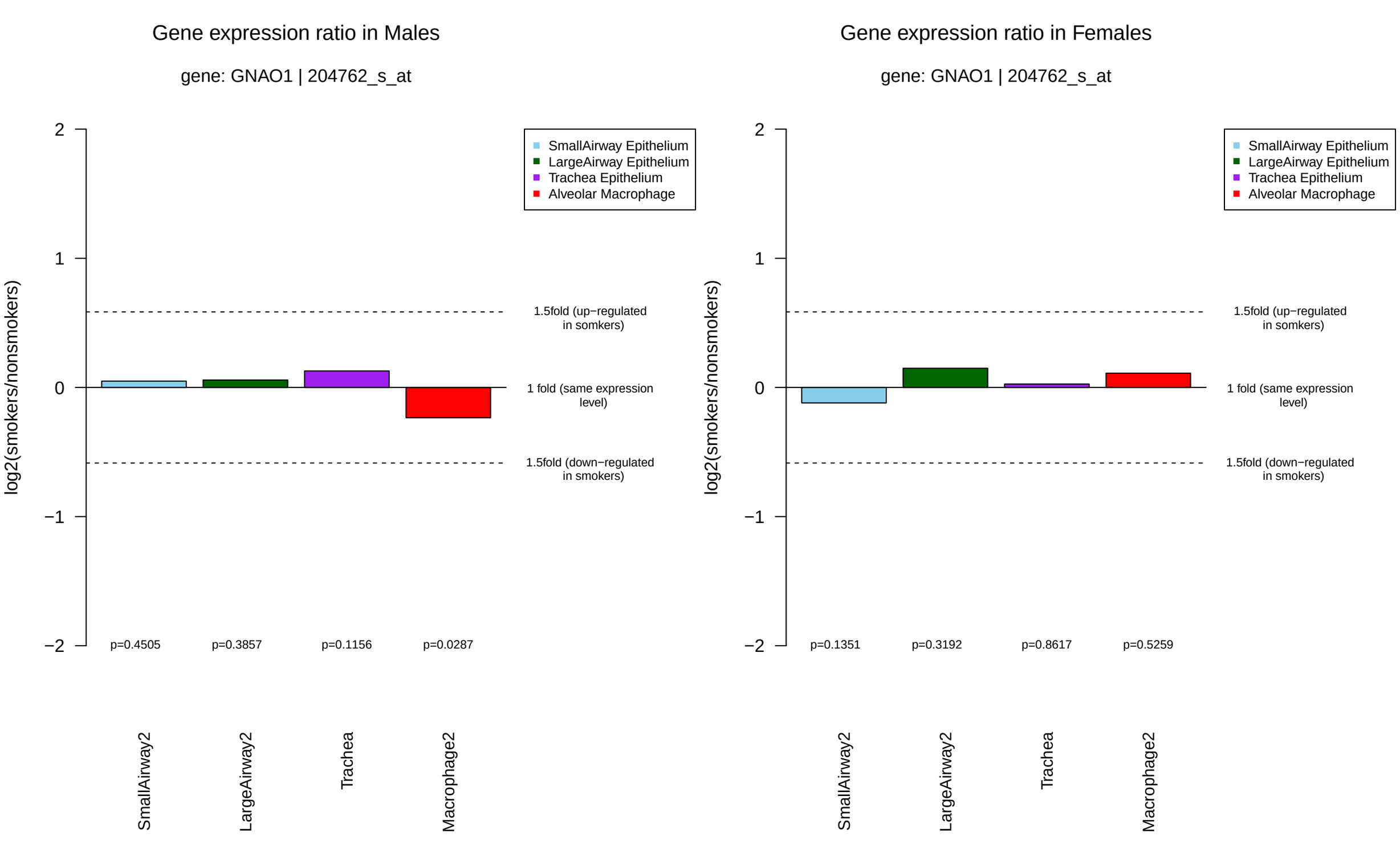
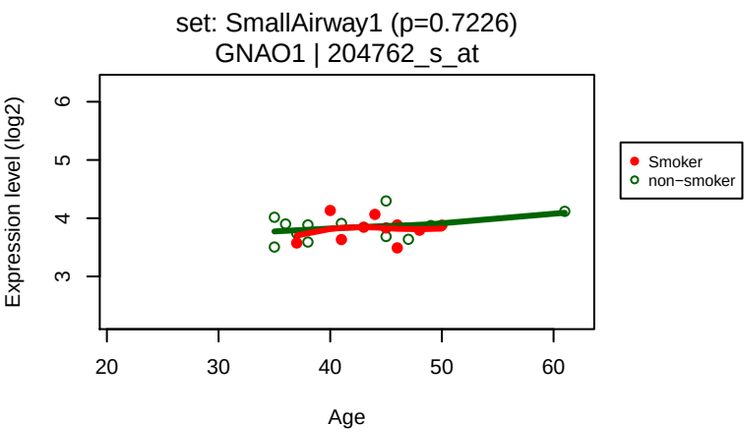
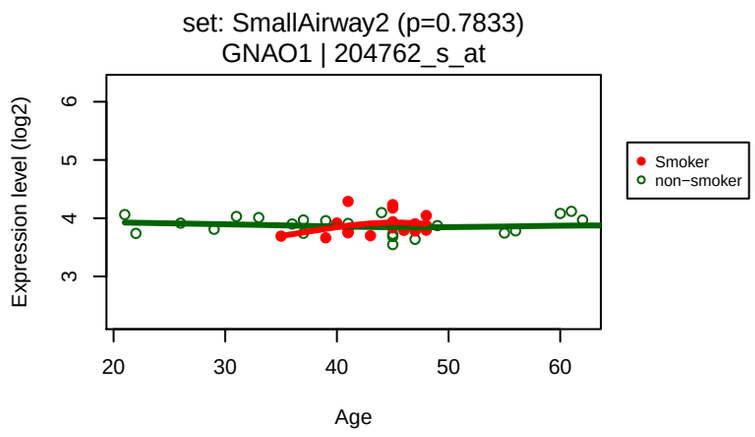
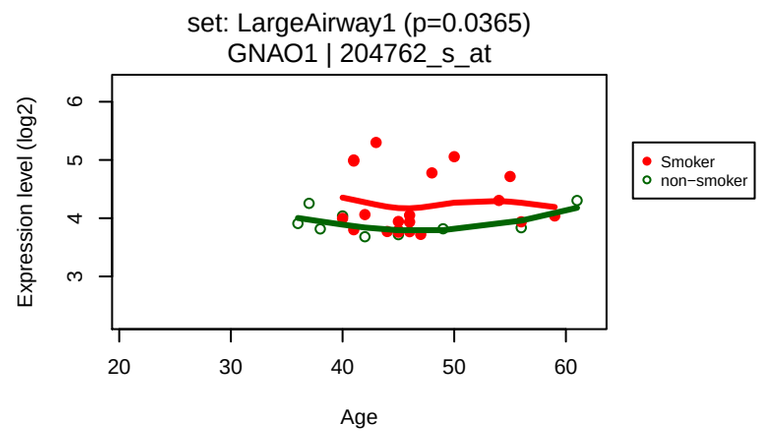

Gene expression of smokers and nonsmokers
gene: GNAO1 | 204762_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7226
p=0.7833
p=0.2053
p=0.0365
p=0.1495
p=0.1718
p=0.8022
p=0.1799
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

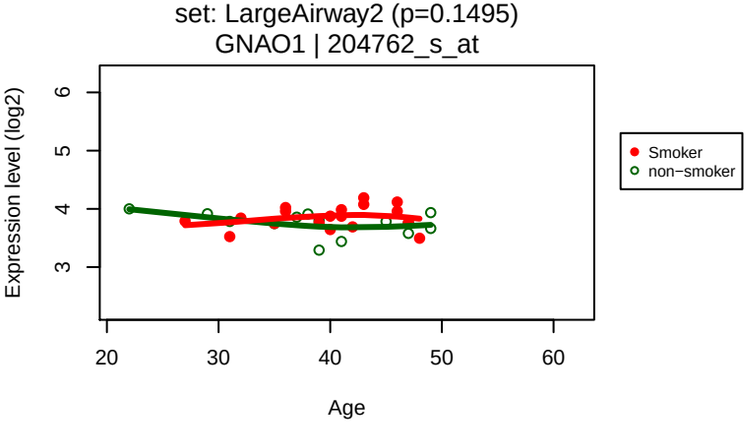

Gene expression ratio (smokers/nonsmokers)
gene: GNAO1 | 204762_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7226
p=0.7833
p=0.2053
p=0.0365
p=0.1495
p=0.1718
p=0.8022
p=0.1799
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

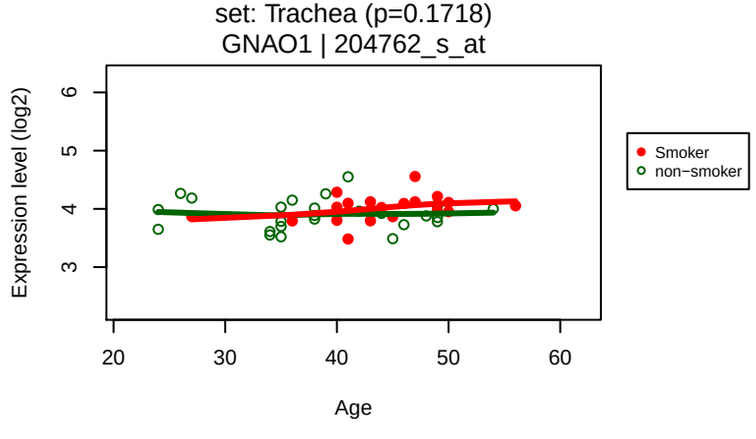

Gene expression ratio in Males
gene: GNAO1 | 204762_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4505
p=0.3857
p=0.1156
p=0.0287
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GNAO1 | 204762_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1351
p=0.3192
p=0.8617
p=0.5259
SmallAirway2
LargeAirway2
Trachea
Macrophage2

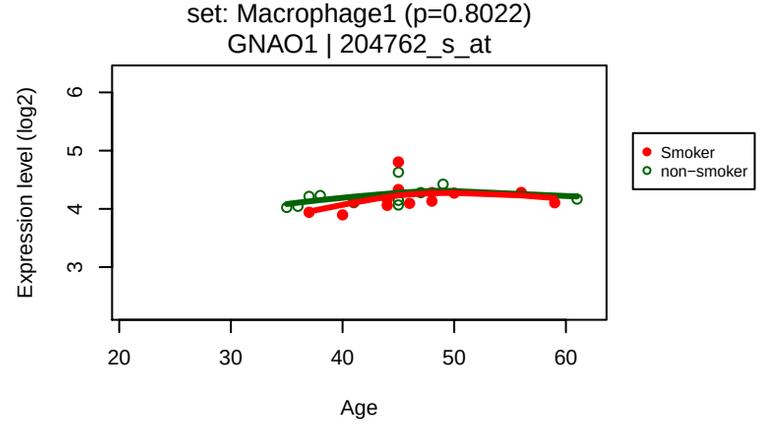

set: SmallAirway1 (p=0.7226)
GNAO1 | 204762_s_at
Expression level (log2)
Age
Smoker
non−smoker

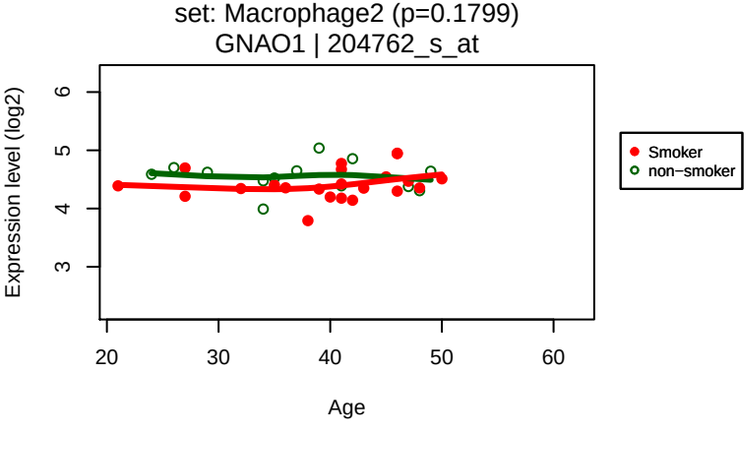

set: SmallAirway2 (p=0.7833)
GNAO1 | 204762_s_at
Expression level (log2)
Age
Smoker
non−smoker

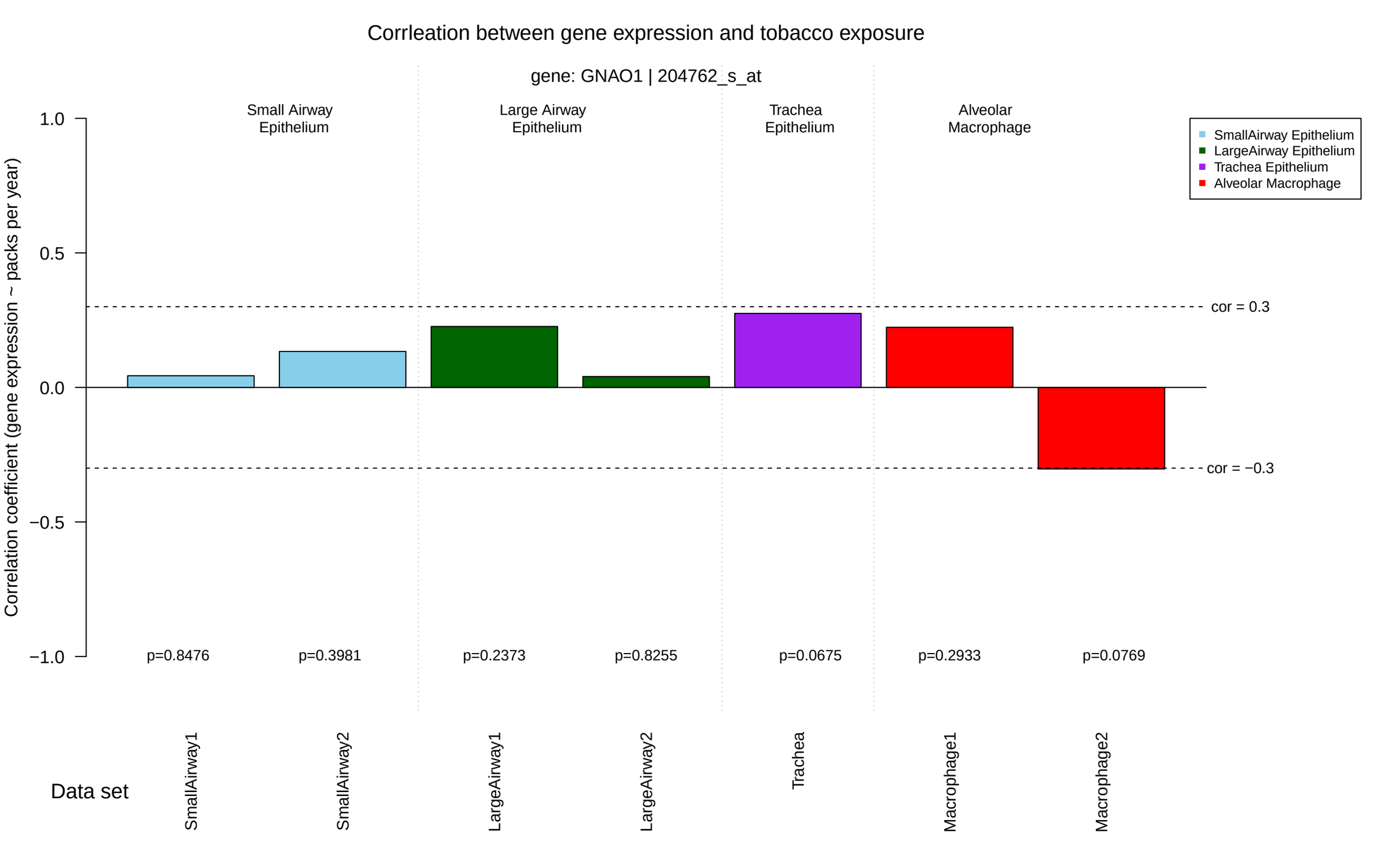

set: LargeAirway1 (p=0.0365)
GNAO1 | 204762_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1495)
GNAO1 | 204762_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1718)
GNAO1 | 204762_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8022)
GNAO1 | 204762_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1799)
GNAO1 | 204762_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GNAO1 | 204762_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8476
p=0.3981
p=0.2373
p=0.8255
p=0.0675
p=0.2933
p=0.0769
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2